$\boxed{\text{Case 2:}}$ $x=y$ $\min(x,y)=x=y=\max(x,y)$

$((x+y)-|x-y|)/2=(x+y-0)/2=x=y$

$((x+y)+|x-y|)/2=(x+y+0)/2=x=y$

$\therefore \min(x,y)=((x+y)-|x-y|)/2$ and $\max(x,y)=((x+y)+|x-y|)/2$

So, for all real numbers x and y, $\min(x,y)=((x+y)-|x-y|)/2$ and $\max(x,y)=((x+y)+|x-y|)/2$ □

8. Prove using the notion of without loss of generality that $5x+5y$ is an odd integer when x and y are integers of opposite parity.

Solution: Let, x and y are two arbitrary integers of opposite parity. Since x and y are arbitrary, the following two cases ① x is odd, y is even ② x is even, y is odd can be combined into a single case.

Let, wlog, x be odd and y be even. By Defn. 1, $\exists$ integers K_1 and K_2 s.t. $x=2K_1+1$ and $y=2K_2$

$\therefore 5x+5y=5(2K_1+1)+5(2K_2)=10K_1+5+10K_2=2(5K_1+2+5K_2)+1$

It's easy to see that $5K_1+2+5K_2$ is an integer. By Defn. 1, $(5x+5y)$ is an odd integer.

9. Prove the triangle inequality, which states that if x and y are real numbers, then $|x|+|y|\geq|x+y|$ (where $|x|$ represents the absolute value of x, which equals x if $x\geq0$ and equals $-x$ if $x<0$).

Solution: Since x and y are real numbers, the following 4 cases arise:

By defn, $|x|\geq0$ and $|y|\geq0$.

$\boxed{\text{Case 1:}}$ $x\geq0, y\geq0$.

Since, $|x|=x$ and $|y|=y$

$\therefore |x|+|y|=x+y=|x+y|\ (\because x+y\geq0)\geq|x+y|$